An Adaptive IMU/UWB Fusion Method for NLOS Indoor Positioning and Navigation

Daquan Feng, *Member, IEEE*, Junjie Peng, Yuan Zhuang, *Senior Member, IEEE*, Chongtao Guo, *Member, IEEE*, Tingting Zhang, *Member, IEEE*, Yinghao Chu, Xiaoan Zhou, and Xiang-Gen Xia, *Fellow, IEEE*

***Abstract*—Indoor positioning system (IPS) plays an important role in the applications of Internet of Things (IoT), including intelligent hospital, logistics, and warehousing. Ultrawideband (UWB)-based IPS has shown superior performance due to its strong multipath resistance and high temporal resolution. However, the non-line-of-sight (NLOS) situations noticeably degrade both the positioning accuracy and the communication reliability. To address this issue, we first propose a support vector machine (SVM)-based channel detection method to distinguish the line-of-sight (LOS) and NLOS conditions. Then, one base station (BS)-based distance and angle positioning algorithm with extended Kalman filter (DAPA-EKF) in NLOS environment is proposed. For the LOS environment, least squares (LSs) with EKF processing of acceleration (LS-AEKF) and velocity (LS-VEKF) are developed. To further improve the performance, the combination of time difference of arrival (TDOA) and KF in LOS environment is proposed. Simulation results show that the positioning accuracy of the proposed algorithm is improved in various environments. Finally, validated using more than 1000 testing positions, the positioning accuracy of LS-AEKF is 73.8%–74.1% higher than that of LS-VEKF among the two proposed algorithms in terms of three or four BSs metrics.**

***Index Terms*—Extended Kalman filter (EKF), indoor positioning system (IPS), inertial measurement unit (IMU), non-line-of-sight (NLOS), ultrawideband (UWB).**

Manuscript received 11 May 2022; accepted 2 February 2023. Date of publication 15 February 2023; date of current version 23 June 2023. This work was supported in part by the National Science and Technology Major Project under Grant 2020YFB1807601; in part by the National Scientific Foundation of China (NSFC) under Grant 62171160; in part by the Fundamental Research Funds for the Central Universities under Grant HIT. OCEF. 2022055; in part by the Excellent Youth Foundation of Hubei Scientific Committee under Grant 2021CFA040; and in part by the Shenzhen Science and Technology Program under Grant JCYJ20210324095209025, Grant JCYJ20190806143212658, Grant JCYJ20190808114213987, and Grant ZDSYS20210623091808025. *(Corresponding author: Chongtao Guo.)*

Daquan Feng is with the Guangdong–Hong Kong Joint Laboratory for Big Data Imaging and Communication and the Guangdong Key Laboratory of Intelligent Information Processing, College of Electronics and Information Engineering, Shenzhen University, Shenzhen 518000, China (e-mail: fdquan@szu.edu.cn).

Junjie Peng is with the West Kowloon Station Police Station, Shenzhen Department, Guangzhou Railway Public Security Bureau, Shenzhen 518001, China (e-mail: 1393724439@qq.com).

Yuan Zhuang is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China (e-mail: yuan.zhuang@whu.edu.cn).

Chongtao Guo is with the Guangdong Key Laboratory of Intelligent Information Processing, College of Electronics and Information Engineering, Shenzhen University, Shenzhen 518000, China (e-mail: ctguo@szu.edu.cn).

Tingting Zhang is with the School of Electronics and Information Engineering, Harbin Institute of Technology, Shenzhen 518055, China, and also with the Broadband Communications Department, Peng Cheng Laboratory, Shenzhen 518066, China (e-mail: zhangtt@hit.edu.cn).

Yinghao Chu is with the Department of Advanced Design and Systems Engineering, City University of Hong Kong, Hong Kong, SAR, China (e-mail: charlie0586@gmail.com).

Xiaoan Zhou is with the College of Electronics and Information Engineering, Shenzhen University, Shenzhen 518000, China (e-mail: xazhou@szu.edu.cn).

Xiang-Gen Xia is with the Department of Electrical and Computer Engineering, University of Delaware, Newark, DE 19716 USA (e-mail: xianggen@udel.edu).

Digital Object Identifier 10.1109/JIOT.2023.3245144

I. Introduction

RECENTLY, high-precision indoor positioning system (IPS) gradually becomes one of the indispensable technologies for the industrial Internet of Things (IoT). With the increasing demands for IPS in numerous applications, different IPS technologies, such as Wi-Fi [1], [2], Bluetooth low energy (BLE) [3], radio frequency identification (RFID) [4], and ultrawideband (UWB) [5], [6], [7] have been proposed and developed. Among these approaches, UWB is one of the most reliable methods due to its centimeter-level accuracy, high transmission rate, low transmission power, and strong penetrating ability. UWB transmits and receives information in extremely narrow pulses of microseconds or less. This narrow pulse communication has high time resolution of the transmitted signal, which has strong anti-interference ability and therefore is essential to achieve high positioning accuracy. Meanwhile, UWB is capable of utilizing wide band of frequency, which indicates strong ability of penetration to reduce the interference of other signals and obstacles. Based on these advantages, many UWB-based methods are proposed for indoor positioning. For instance, two way ranging (TWR) algorithm is proposed based on UWB measured distances [8]. Then, the measured distances can be used by the least square (LS) positioning algorithm to obtain the target position [9]. In [10], double-sided TWR (DS-TWR) algorithm is proposed to eliminate the need for additional clock synchronization mechanisms. To address the problem of frequency drift caused by crystal clock drift, an enhanced asymmetric DS-TWR (EADS-TWR) algorithm is proposed [11]. Coordination positioning system is often used to balance the cost, latency, and reliability in positioning [12]. A positioning system based on the direction-of-arrival (DOA) estimation of signals is proposed [13]. Angle of arrival (AOA) [14], time of arrival (TOA) [15], time difference of arrival (TDOA) [16], [17], and

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radio signal strength (RSS) [18] are also commonly used for positioning. In addition, fusion-based methods are developed by combining UWB with other methods to further improve the positioning accuracy. For instance, UWB is combined with inertial measurement unit (IMU) through extended Kalman filter (EKF) in [19] and [20]. These methods can achieve high-precision estimations of position and orientation with low cost. However, the accuracy of UWB-based methods in real applications may be degraded due to the non-line-of-sight (NLOS) issue [21], [22].

Many methods have been proposed to address the NLOS issue. The key idea is to detect the NLOS situation and then mitigate the negative effects of NLOS. To detect the NLOS condition, methods include but not limited to multipath channel statistics [23], [24], distance estimation based on the first and multiple paths [25], [26], model-based approach by analysing the channel impulse response (CIR) [27], [28], and data driven approach, such as support vector machine (SVM), multilayer perception (MLP), decision trees, and other machine-learning methods [29], [30], [31], [32]. In addition, the Rician K-factor, defined as the ratio of line-of-sight (LOS) and NLOS component powers, can also be used to identify NLOS. However, random shadowing effects in mobile communications may cause fluctuating LOS component, making the performance of the Rician K-factor-based approach degraded [33]. For comprehensive discussion and comparison among different NLOS detection methods, one can refer to [34] further. Based on the detected NLOS conditions, some filtering algorithms are used to mitigate the errors. An eight-state EKF is proposed in [21], where a 2-D positioning system is considered. However, the state information is complex and the calculation of observation noise is heavy. When the base station (BS) signal is lost, it will not be able to locate accurately. In [35], a multisensor fusion is proposed to obtain the region of interest based on a vision sensor, and EKF fusion positioning is adopted to combine UWB and IMU measurement. Although multiple sensors can supplement the signal loss in the NLOS scene, too many sensors will increase the computation cost. Besides, the errors between sensors will affect each other and degrade the positioning performance. In [22], a loose-coupling EKF algorithm is proposed. The observation vector adopted is the deviation of each measured value from the actual value. However, the actual value is difficult to obtain, making its applications limited. In [36], two-step KF is proposed to linearize the nonlinear problem. Because the acceleration is added to a certain white Gaussian noise in the simulation, the two-step KF can effectively reduce the error. However, the acceleration noise in practical systems is not ideal. The state-of-the-art IPS researches may not meet the requirements of scene switching in practical applications. In order to develop an NLOS detection method that is as simple as possible and can adapt to different scenarios, it is necessary to reduce the complexity and the deployment cost of positioning BS, and address positioning signal loss or low positioning accuracy in NLOS scenarios.

In this article, an adaptive fusion method with improved NLOS detection and positioning algorithm is proposed and validated to address the aforementioned issues. This algorithm first derives proportional features between the UWB measured distance and the actual distance under different channels, and then employs an SVM to learn and to detect whether the NLOS situation is present. Afterward, based on the number of BSs available under LOS, the proposed algorithm adaptively employs different optimized algorithms to conduct the positioning. Specifically, when one or two BSs are available, a distance and angle positioning algorithm EKF is adopted, which is denoted as distance and angle positioning algorithm with EKF (DAPA-EKF) in this work. When three or four BSs are available, two algorithms based on LS with acceleration EKF (LS-AEKF) and velocity EKF (LS-VEKF) are proposed, as well as the one based on TDOA with Kalman filter positioning (TDOA-KF). Compared with [21], this article adopts symmetrical and bilateral transmission mode on the basis of TWR to improve the ranging accuracy; compared with [5] and [11], NLOS recognition and detection are increased. The proposed fusion algorithm is suitable for indoor complex environment, improves the utilization of sensor data, and effectively reduces the positioning offset caused by either acceleration or velocity noise.

The main contributions of this work are exhibited as follows.

1) We propose a low-complexity SVM-based NLOS identification method using ranging statistics information. This proposed method provides a good solution to many practical systems, where massive sensors have strict real-time indoor positing requirement.
2) We develop a new sensor fusion strategy called DAPA-EKF that works well even when the number of BS in LOS is less than three. Assisted by the distance information provided by the UWB sensors, the positioning accuracy of the IMU can be enhanced, which extends the application to the case of insufficient available BSs.
3) We design the sensor fusion strategy called LS-AEKF and LS-VEKF that lead to better positioning accuracy than that of the baseline schemes. The proposed algorithms are more robust to sensor information error since the fusion of different sensors introduces the diversity gain. Numerical results show that the proposed algorithms outperform the baseline methods when three or four BSs are in LOS.
4) In addition to software simulation, we build a prototype of the positioning system in the lab to evaluate its performance. Experimental results show that the proposed algorithm achieves superior positioning accuracy over multiple reference methods.

This article is organized as follows. In Section II, we present the principle of UWB and IMU positioning. The detection of NLOS and the fusion positioning strategy of UWB and IMU are discussed in Section III. Then, we show simulation and experimental results in Section IV. Finally, the conclusions are drawn in Section V.

II. Building Modules of the Adaptive Fusion Positioning Method

In this section, we first describe the adaptive fusion positioning framework, and then introduce its key building blocks,

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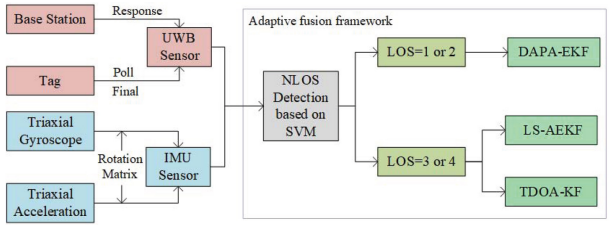


Fig. 1. Schematic of the adaptive positioning method.

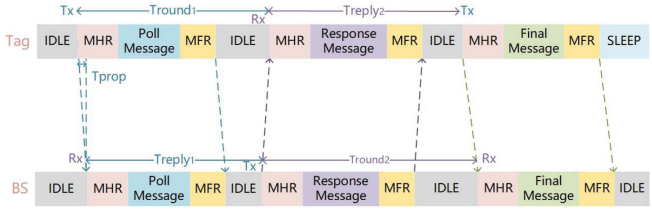


Fig. 2. SDS-TWR ranging principle.

including UWB and IMU systems. As shown in Fig. 1, the proposed method receives inputs from both UWB and IMU sensors. Particularly, UWB sensors obtain the distance information, and IMU sensors obtain acceleration and velocity information. Then, the SVM-based method is adopted to evaluate the distance information calculated by UWB to perform NLOS detection. Eventually, based on the derived number of BSs under LOS, the fusion algorithm uses the optimal strategy accordingly to perform the positioning.

A. UWB Module

UWB is a low-power wireless technology and transmits data through nanosecond pulses. With a data transmission rate about hundreds of megabits per second, UWB is suitable for high-speed, short-distance wireless personal communication. In our scheme, UWB is used for measuring distance.

Fig. 2 shows the communication process of the symmetric double sided TWR (SDS-TWR) positioning mode in UWB. The tag transmits (Tx) poll message when it is idle, and the BS receives (Rx) the signal after the transmission time, Tprop. After the BS is idle, it transmits the response message to the tag. Then, after the tag receives the response message, it will be idle for a period of time and then transmit the final message. The times of the tag from transmitting to receiving and from receiving to transmitting are Tround_1 and Treply_2, respectively. The times of the BS from receiving to transmitting and from transmitting to receiving are Treply_1 and Tround_2, respectively. The transmitted message frame includes frame control domain, beacon sequence number, and addressing information domain (MHR), load, and frame check sequence of 16 bits (MFR).

B. IMU Module

IMU system consists of gyroscope, accelerometer, and magnetometer, which can estimate the 3-D position, speed, and direction of the target [37]. The positioning is conducted using the direction and acceleration information. Due to the cumulative error and temperature deviation in the angle integration

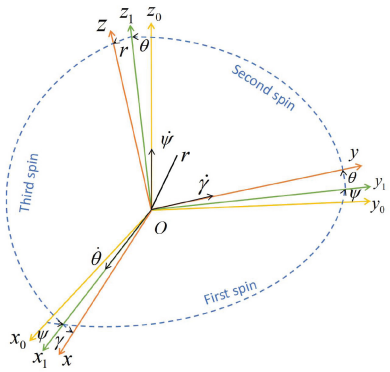


Fig. 3. Transformation of coordinate system.

calculation, the accuracy of IMU positioning decreases with time. The inertial navigation coordinate system is established based on Newtonian mechanics. The navigation coordinate system is obtained by rotating the carrier coordinate system where the tag is located as shown in Fig. 3.

The coordinate system $Ox_0y_0z_0$ can be converted to $Oxyz$ by three rotations along the axes. First, the z_0-axis is rotated angle ψ to $Ox_1y_1z_0$. Then, the x_1-axis is rotated angle θ to Ox_1yz_1. Finally, the y-axis is rotated angle γ to $Oxyz$ [11].

Vector $\boldsymbol{r}$ in the coordinate system $Ox_0y_0z_0$ can be projected to the coordinate system $Ox_1y_1z_0$ as

$$\begin{bmatrix} \boldsymbol{r}_{x_1} \\ \boldsymbol{r}_{y_1} \\ \boldsymbol{r}_{z_0} \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{r}_{x_0} \\ \boldsymbol{r}_{y_0} \\ \boldsymbol{r}_{z_0} \end{bmatrix} \tag{1}$$

abbreviated as

$$\boldsymbol{r}_1 = \boldsymbol{C}_0^1 \boldsymbol{r}_0. \tag{2}$$

After the second rotation of θ, the projection of vector $\boldsymbol{r}$ in the coordinate system Ox_1yz_1 is

$$\begin{bmatrix} \boldsymbol{r}_{x_1} \\ \boldsymbol{r}_{y} \\ \boldsymbol{r}_{z_1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \boldsymbol{r}_{x_1} \\ \boldsymbol{r}_{y_1} \\ \boldsymbol{r}_{z_0} \end{bmatrix} \tag{3}$$

abbreviated as

$$\boldsymbol{r}_2 = \boldsymbol{C}_1^2 \boldsymbol{r}_1. \tag{4}$$

Then, the $Oxyz$ coordinate system is obtained by the third rotation, and the projection of vector $\boldsymbol{r}$ in $Oxyz$ coordinate system can be expressed as

$$\begin{bmatrix} \boldsymbol{r}_{x} \\ \boldsymbol{r}_{y} \\ \boldsymbol{r}_{z} \end{bmatrix} = \begin{bmatrix} \cos\gamma & 0 & -\sin\gamma \\ 0 & 1 & 0 \\ \sin\gamma & 0 & \cos\gamma \end{bmatrix} \begin{bmatrix} \boldsymbol{r}_{x_1} \\ \boldsymbol{r}_{y} \\ \boldsymbol{r}_{z_1} \end{bmatrix} \tag{5}$$

abbreviated as

$$\boldsymbol{r}_3 = \boldsymbol{C}_2^3 \boldsymbol{r}_2 = \boldsymbol{C}_2^3 \boldsymbol{C}_1^2 \boldsymbol{C}_0^1 \boldsymbol{r}_0 = \boldsymbol{C}_0^3 \boldsymbol{r}_0. \tag{6}$$

Thus, the rotation matrix from the carrier coordinate system to the platform coordinate system can be obtained

$$\boldsymbol{C}_0^3 = \begin{bmatrix} \cos\psi\cos\gamma - \sin\psi\sin\gamma\sin\theta & \cos\gamma sin\psi - \sin\gamma\sin\theta\cos\psi & -\sin\gamma\cos\theta \\ -\cos\theta\sin\psi & \cos\psi\cos\theta & \sin\theta \\ \sin\gamma\cos\psi + \cos\gamma\sin\psi\sin\theta & \sin\gamma\sin\psi - \cos\gamma\sin\theta\cos\psi & \cos\gamma\cos\theta \end{bmatrix}. \tag{7}$$

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Considering that θ, γ, and ψ are small, therefore the second order components can be neglected and the equation can be simplified as

$$\boldsymbol{C}_0^3 = \begin{bmatrix} 1 & \psi & -\gamma \\ -\psi & 1 & \theta \\ \gamma & -\theta & 1 \end{bmatrix}. \tag{8}$$

The geographic coordinate system is selected as the navigation coordinate system, and the velocity and position equations are obtained from the inertial navigation basic formula

$$\dot{\boldsymbol{v}}^n = \boldsymbol{C}_b^n \boldsymbol{f}^b + \boldsymbol{g}^n - \left(\boldsymbol{\omega}_{en}^n + 2\boldsymbol{\omega}_{ie}^n\right) \times \boldsymbol{v}^n \tag{9}$$

$$\dot{\boldsymbol{P}} = \boldsymbol{R}_c \boldsymbol{v}^n \tag{10}$$

where $\boldsymbol{v}^n$ is the projection of the velocity of the carrier relative to the earth on the navigation coordinates, $\boldsymbol{C}_b^n$ is the attitude matrix to convert navigation coordinate system into the carrier coordinate system, $\boldsymbol{f}^b$ is the specific force acceleration obtained by the accelerometer in the carrier coordinate system, $\boldsymbol{\omega}_{ie}^n$ and $\boldsymbol{\omega}_{en}^n$ are the projection of the rotation angular velocity of the earth coordinate system relative to the inertial coordinate system, $\boldsymbol{\omega}_{en}^n$ is the navigation coordinate system relative to the geographical coordinate system, $\boldsymbol{P} = [\lambda, \phi, h]^T$ is the position, $\boldsymbol{R}_c$ is the local curvature matrix, and H is speed update cycle. By integrating the velocity, the target position can be calculated as

$$\begin{aligned} \boldsymbol{r}(t_{k+1}) = \boldsymbol{C}_{n(t_k)}^{n(t_{k+1})} \Bigg[& H\boldsymbol{v}^n(t_k) + \boldsymbol{C}_{b(t_k)}^{n(t_k)} \int_{t_k}^{t_{k+1}} \int_{t_k}^{t} \boldsymbol{C}_{b(\tau)}^{b(t_k)} \boldsymbol{f}^b \\ & d\tau dt + \int_{t_k}^{t_{k+1}} \boldsymbol{C}_{n(t)}^{n(t_k)} \boldsymbol{\omega}_{in}^n \times \boldsymbol{r}^n dt - \int_{t_k}^{t_{k+1}} \int_{t_k}^{t} \boldsymbol{C}_{n(\tau)}^{n(t_k)} \\ & \boldsymbol{\omega}_{ie}^n \times \boldsymbol{v}^n d\tau dt + \int_{t_k}^{t_{k+1}} \int_{t_k}^{t} \boldsymbol{C}_{n(\tau)}^{n(t_k)} \boldsymbol{g}^n d\tau dt \Bigg]. \end{aligned} \tag{11}$$

This equation is the position integral equation with the consideration of the navigation coordinate rotation. In addition, the second term in the square bracket is the position increment caused by the second integral of the acceleration of proportional–integral in the time interval. To achieve a high-precision navigation coordinate system, the scroll compensation term caused by the high-frequency vibration of the carrier must be considered. In summary, the position $\boldsymbol{r}(t_{k+1})$ is composed of the increment caused by velocity, proportional–integral, acceleration, and angular velocity.

III. Adaptive Fusion Positioning System

In this section, we first introduce the NLOS detection method based on SVM, and then propose the adaptive fusion positioning schemes according to the detected channel conditions. The positioning state is named according to the number of BSs in LOS in the experimental scene. LOS is set as LOS $= m$ when m BSs are in LOS, where at most four BSs are considered in this article. When LOS $= 1$ or 2, DAPA-EKF algorithm is proposed and verified by simulation. When LOS $= 3$ or 4, an LS-AEKF algorithm is proposed, and a DAPA-EKF algorithm is used to optimize the results.

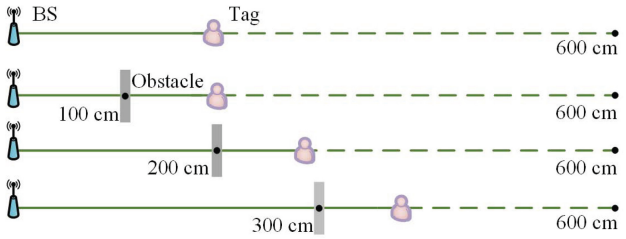


Fig. 4. Sample collection control group experiment.

A. SVM-Based NLOS Detection

The accuracy of ranging is seriously affected by NLOS channels, so it is necessary to identify the channel condition before positioning. The ranging accuracy of SDS-TWR mainly depends on the responding and processing time of the BS, as well as the frequency accuracy between the BS and the tag. Although UWB has both strong anti-interference and anti-multipath ability, NLOS situations, such as moving obstacles, may adversely affect the ranging accuracy of UWB system [38], [39]. Therefore, the implementation of NLOS detection is a prerequisite for mitigation of positioning error. Since the range statistics-based approach usually has much lower hardware and software complexity while achieving fairly good performance [34], [40], we resort to the range statistics information to identify NLOS. With the fact that NLOS identification problem is actually a binary classification issue in mathematics, it is more appropriate to utilize the SVM method rather than other machine-learning approaches [41], [42]. The reason lies in two points. First, SVMs are supervised learning models that analyze data for classification and regression, which are generally easier to implement and lower in complexity than other machine-learning-based methods. Second, the range statistics information usually has much lower dimension of characteristics than channel characteristics, making the SVM models be qualified enough, while sophisticated machine-learning methods, e.g., deep neural network, could bring about more complexity in model training.

We set the obstacles as pedestrians since NLOS is usually caused by pedestrians in the indoor scene. In order to fully consider the impact of obstacles on ranging, different distances of obstacles are set for measurement. Four different control experiments are illustrated in Fig. 4. First, we measure the UWB distance and the actual distances when the tag is moving without obstruction. Then control experiments with obstacles are set up as follows: 1) fixing the position of the obstacles at 1/6, 1/3, and 1/2 of the total test length; 2) moving the tag blocked by the obstacles; 3) recording the actual and measured distance of each point; and 4) comparing the errors at different distances from the obstacles. Fig. 5 shows the result of the ratio of the measured value and the actual value. When the BSs are too close to each other, it will cause serious multipath effect, so the error is large in the initial short distance. It can be seen from the figure that obstacles in different positions have different effects on ranging. But with the increase of tag distance, the measurement error is within a certain range, so the ratio of the distance to the actual distance gradually decreases. Therefore, an SVM model can be trained by ratio

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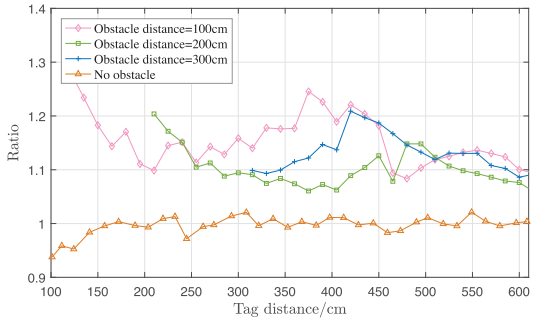


Fig. 5. Ratio of measured value to actual value.

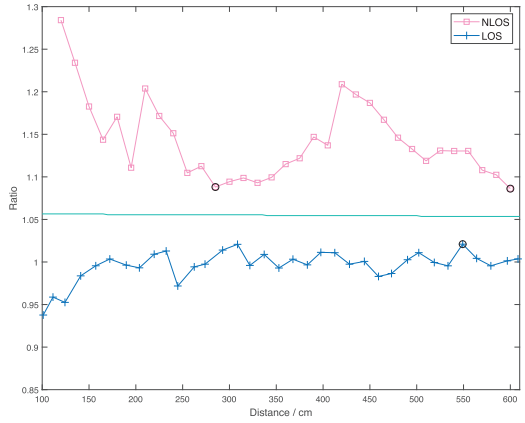


Fig. 6. SVM classification results.

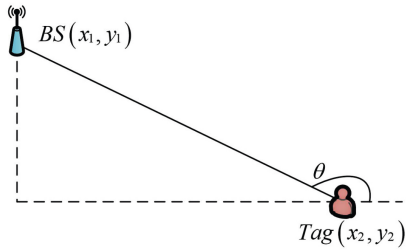


Fig. 7. Positioning diagram for LOS = 1.

data in the presence and absence of obstacles to detect NLOS situations.

To collect the data for SVM model training, we locate a tag at N points equidistantly between 100 and 600 cm from a BS. The measured distance and actual distance covered by obstacles are d_1 and l_1, respectively. The positive point $([d_1^i/l_1^i], +1)$, $i = 1, 2, \ldots, N$, in the NLOS environment is obtained by dividing the ith measured value d_1^i by the actual distance l_1^i. The measured distance and actual distance in the absence of obstacles are d_2 and l_2, respectively. The negative point $([d_2^i/l_2^i], -1)$, $i = 1, 2, \ldots, N$, in the LOS environment is obtained by dividing the ith measurement value d_2^i by the actual distance l_2^i. With the data mentioned above, the SVM model finds a hyperplane to effectively classify NLOS and LOS situations. This hyperplane is defined using a linear system of equations

$$\boldsymbol{\omega}^T \boldsymbol{x} + b = 0. \tag{12}$$

$\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots, \omega_d)$ is the normal vector, which determines the direction of the hyperplane, and b is the displacement term, which determines the distance between the hyperplane and the origin. Considering the training samples can be classified by the hyperplane, we have $\boldsymbol{\omega}^T \boldsymbol{x}_i + b \geq +1$ if the value of sample point y_i equals $+1$, and $\boldsymbol{\omega}^T \boldsymbol{x}_i + b \leq -1$ if y_i equals -1, i.e.,

$$\begin{cases} \boldsymbol{\omega}^T \boldsymbol{x}_i + b \geq +1, \ y_i = +1 \\ \boldsymbol{\omega}^T \boldsymbol{x}_i + b \leq -1, \ y_i = -1. \end{cases} \tag{13}$$

The support vector is the closest sample to the hyperplane that satisfies the above equations.

The training results of the SVM model are shown in Fig. 6. The positive predictions from the SVM are marked as pink squares while the negative predictions from the SVM are marked as blue cross. The line in the center is the SVM hyperplane. It can be seen from the results that our proposed SVM model effectively separates the NLOS instances from the LOS instances. In real-time applications, the threshold value will be updated with collected data. For example, the IMU uses the current state to calculate the position of the next moment, so that the distance between the IMU derived position and the BS is obtained as the actual distance, which is then used with the UWB predicted distance to calculate the input ratio.

B. DAPA-EKF Positioning Algorithm for LOS = 1 or 2

As shown in Fig. 7, when there is only one BS signal under LOS, it is denoted as LOS = 1. In this case, one BS-based distance angle positioning algorithm (DAPA) is proposed. Since a single BS cannot obtain the movement direction of the tag, and IMU can only calculate the direction change value of the tag during movement, this method needs to know the initial movement direction of the tag. When there is ranging information from at least three BSs, the current accurate position and initial direction can be obtained.

The positioning equation of DAPA is shown below. Tag location (x_2, y_2) is calculated based on the known BS location (x_1, y_1), angle θ obtained by IMU integration, and distance d measured by UWB, where θ and d are given by

$$\begin{cases} \theta = \arctan\frac{y_2 - y_1}{x_2 - x_1} \\ d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \end{cases} \tag{14}$$

Due to the observation errors of the measured distance and angle, the position error of the solution is relatively large. EKF can effectively estimate the parameters of the nonlinear system. Based on IMU, the state transition matrix $\boldsymbol{\Phi}$ and noise driven matrix $\boldsymbol{\Gamma}$ can be obtained in (15). We set the white noise to $\boldsymbol{W}(k) = \begin{bmatrix} w_x(k) & w_y(k) \end{bmatrix}^T$ with a mean value of zero and covariance matrix $\boldsymbol{Q} = \text{diag}(\sigma_{a_x}^2, \sigma_{a_y}^2)$

$$\boldsymbol{\Phi} = \begin{bmatrix} 1 & 0 & T & 0 & \frac{T^2}{2} & 0 \\ 0 & 1 & 0 & T & 0 & \frac{T^2}{2} \\ 0 & 0 & 1 & 0 & T & 0 \\ 0 & 0 & 0 & 1 & 0 & T \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \boldsymbol{\Gamma} = \begin{bmatrix} \frac{T^3}{6} & 0 \\ 0 & \frac{T^3}{6} \\ \frac{T^2}{2} & 0 \\ 0 & \frac{T^2}{2} \\ T & 0 \\ 0 & T \end{bmatrix}. \tag{15}$$

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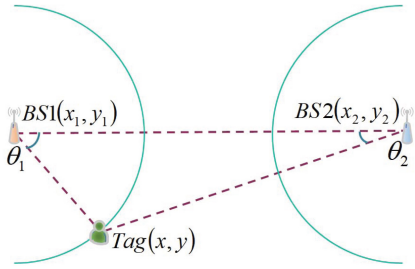


Fig. 8. Positioning diagram for LOS = 2.

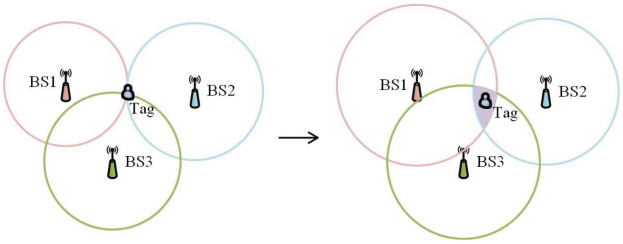


Fig. 9. Positioning diagram for the LS method.

We set the state vector to $\boldsymbol{X}(k) = \begin{bmatrix} x_x(k) & x_y(k) \end{bmatrix}^T$, observation noise $\boldsymbol{V}(k) = \begin{bmatrix} v_\theta(k) & v_d(k) \end{bmatrix}^T$ with a mean value of zero and covariance matrix $\boldsymbol{R} = \text{diag}(\sigma_\theta, \sigma_d)$. $\hat{\theta}(k)$, and $\hat{d}$ are the predicted angle and distance, respectively. The observation vector is

$$\boldsymbol{Z}(k) = \begin{bmatrix} \hat{\theta}(k) \\ \hat{d}(k) \end{bmatrix} + \boldsymbol{V}(k). \tag{16}$$

The observation matrix $\boldsymbol{H}(k)$ is set based on Jacobian matrix. x_{BS} and y_{BS} represent the position of the BS. Set the observation equation as

$$\boldsymbol{Z}(k) = \boldsymbol{H}(k)\boldsymbol{X}(k) + \boldsymbol{V}(k) \tag{17}$$

where

$$\boldsymbol{H}(k) = \begin{bmatrix} -\frac{x_y(k) - y_{\text{BS}}}{\hat{d}(k)^2} & \frac{x_x(k) - x_{BS}}{\hat{d}(k)^2} \\ \frac{x_x(k) - x_{\text{BS}}}{\hat{d}(k)} & \frac{x_y(k) - y_{\text{BS}}}{\hat{d}(k)} \end{bmatrix}. \tag{18}$$

Based on EKF [43], [44], [45], the updated position can be obtained.

As shown in Fig. 8, when LOS = 2, we can measure the distances d_1, d_2, and angles θ_1, θ_2 between the tag and BS1(x_1, y_1) and BS2(x_2, y_2), respectively. A DAPA-EKF positioning algorithm based on two BSs is proposed to reduce measurement errors and to improve positioning accuracy through EKF. The distance from BS2 to tag in the figure is farther than that of BS1 in Fig. 8, and the corresponding angle error is smaller. Therefore, θ_2 is selected as the positioning angle for the scene shown in the figure. The equations for solving the tag position are

$$\begin{cases} |\sqrt{(x_1 - x)^2 + (y_1 - y)^2} - \sqrt{(x_2 - x)^2 + (y_2 - y)^2}| = |d_1 - d_2| \\ \tan(\theta_2) = \frac{y_2 - y}{x_2 - x}. \end{cases} \tag{19}$$

We set the state vector as $\boldsymbol{X}(k) = \begin{bmatrix} x_x(k) & x_y(k) \end{bmatrix}^T$, and observation noise $\boldsymbol{V}(k) = \begin{bmatrix} v_{\theta_1}(k) & v_{\theta_2}(k) & v_{d_1}(k) & v_{d_2}(k) \end{bmatrix}^T$ with a mean value of zero covariance matrix $\boldsymbol{R} = \text{diag}(\sigma_{\theta_1}, \sigma_{\theta_2}, \sigma_{d_1}, \sigma_{d_2})$. $\hat{\theta}_1(k), \hat{\theta}_2(k), \hat{d}_1(k)$, and $\hat{d}_2(k)$ are the predicted angles and distances. The observation vector is

$$\boldsymbol{Z}(k) = \begin{bmatrix} \hat{\theta}_1(k) \\ \hat{\theta}_2(k) \\ \hat{d}_1(k) \\ \hat{d}_2(k) \end{bmatrix} + \boldsymbol{V}(k). \tag{20}$$

The observation matrix $\boldsymbol{H}(k)$ is set based on Jacobian matrix, x_{BS} and y_{BS}, respectively, represent coordinates of the BS on the x- and y-axis. We set the observation equation as

$$\boldsymbol{Z}(k) = \boldsymbol{H}(k)\boldsymbol{X}(k) + \boldsymbol{V}(k) \tag{21}$$

where

$$\boldsymbol{H}(k) = \begin{bmatrix} -\frac{x_y(k) - y_{\text{BS}_1}}{\hat{d}_1(k)^2} & \frac{x_x(k) - x_{\text{BS}_1}}{\hat{d}_1(k)^2} \\ -\frac{x_y(k) - y_{\text{BS}_2}}{\hat{d}_2(k)^2} & \frac{x_x(k) - x_{\text{BS}_2}}{\hat{d}_2(k)^2} \\ \frac{x_x(k) - x_{\text{BS}_1}}{\hat{d}_1(k)} & \frac{x_y(k) - y_{\text{BS}_1}}{\hat{d}_1(k)} \\ \frac{x_x(k) - x_{\text{BS}_2}}{\hat{d}_2(k)} & \frac{x_y(k) - y_{\text{BS}_2}}{\hat{d}_2(k)} \end{bmatrix}. \tag{22}$$

Based on the observation vector and observation matrix by EKF algorithm flow, we can update the state matrix and covariance matrix.

C. LS-AEKF Positioning Algorithm for LOS ≥ 3

When LOS ≥ 3, UWB uses LS to locate the tag based on the distance information. As shown in Fig. 9, under the influence of the observation noise, the positioning result is in the intersection area of the circles centered on the BSs. To further reduce the ranging error, an LS-AEKF and an LS-VEKF positioning algorithms based on LS are proposed after the EKF processing of acceleration and velocity, respectively. First, the LS algorithm for calculating the tag coordinates is defined based on the distance d_n from the tag to BSn (x_n, y_n), as shown in (22)

$$\begin{cases} (x_1 - x)^2 + (y_1 - y)^2 = d_1^2 \\ (x_2 - x)^2 + (y_2 - y)^2 = d_2^2 \\ \vdots \\ (x_n - x)^2 + (y_n - y)^2 = d_n^2. \end{cases} \tag{23}$$

Subtracting the last equation from all equations, we have

$$\boldsymbol{A} = -2 \begin{bmatrix} x_1 - x_n & y_1 - y_n \\ x_2 - x_n & y_2 - y_n \\ \vdots & \vdots \\ x_{n-1} - x_n & y_{n-1} - y_n \end{bmatrix} \tag{24}$$

$$\boldsymbol{b} = \begin{bmatrix} d_1^2 - d_n^2 + x_n^2 + y_n^2 - x_1^2 - y_1^2 \\ d_2^2 - d_n^2 + x_n^2 + y_n^2 - x_2^2 - y_2^2 \\ \vdots \\ d_{n-1}^2 - d_n^2 + x_n^2 + y_n^2 - x_{n-1}^2 - y_{n-1}^2 \end{bmatrix}. \tag{25}$$

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With the above equations, we can derive the coordinate of the tag as

$$\boldsymbol{X} = \left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\boldsymbol{b} \tag{26}$$

where the coordinate matrix of the tag is $\boldsymbol{X} = [x \quad y]^T$. To minimize the observation noise of distance prediction, the positioning accuracy can be improved by integrating the positioning information of IMU with EKF. In addition, two different positioning strategies are discussed. One strategy considers only velocity and the other considers both velocity and acceleration. The LS-AEKF positioning method, including acceleration and velocity information is shown as follows. Defining the state vector as $\boldsymbol{X}(k) = \left[x_x(k) \quad x_y(k)\right]^T$, the observation noise vector $\boldsymbol{V}(k) = \left[v_{d_1}(k) \quad v_{d_2}(k) \quad v_{d_3}(k)\right]^T$ with a mean value of zero covariance matrix $\boldsymbol{R} = \text{diag}(\sigma_{d_1}, \sigma_{d_2}, \sigma_{d_3})$, $\hat{d}_1(k), \hat{d}_2(k)$, and $\hat{d}_3(k)$ are the predicted distances calculated according to the predicted coordinates.

The IMU predicts the position of the next moment based on the current position, acceleration, velocity, and direction information. The estate transition matrix $\boldsymbol{\Phi}$ and noise driven matrix $\boldsymbol{\Gamma}$ are the same as formula (15). Therefore, the predicted distance is obtained by calculating the distance from the predicted location to the BSs

$$\boldsymbol{Z}(k) = \begin{bmatrix} \hat{d}_1(k) \\ \hat{d}_2(k) \\ \hat{d}_3(k) \end{bmatrix} + \boldsymbol{V}(k). \tag{27}$$

The observation matrix $\boldsymbol{H}(k)$ is set based on Jacobian matrix. Set the observation equation as

$$\boldsymbol{Z}(k) = \boldsymbol{H}(k)\boldsymbol{X}(k) + \boldsymbol{V}(k) \tag{28}$$

where

$$\boldsymbol{H}(k) = \begin{bmatrix} \frac{x_x(k)-x_{\text{BS}_1}}{\hat{d}_1} & \frac{x_y(k)-y_{\text{BS}_1}}{\hat{d}_1} \\ \frac{x_x(k)-x_{\text{BS}_2}}{\hat{d}_2} & \frac{x_y(k)-y_{\text{BS}_2}}{\hat{d}_2} \\ \frac{x_x(k)-x_{\text{BS}_3}}{\hat{d}_3} & \frac{x_y(k)-y_{\text{BS}_3}}{\hat{d}_3} \end{bmatrix} \tag{29}$$

where x_{BS} and y_{BS} characterizes the position of the BS.

If the IMU acceleration is not considered, the positioning of IMU depends on the velocity information and the predicted position is calculated by

$$\begin{cases} x_x(k+1) = x_x(k) + v_x(k)T \\ x_y(k+1) = x_y(k) + v_y(k)T. \end{cases} \tag{30}$$

Different from LS-AEKF, only velocity and direction information are used in LS-VEKF, without considering the acceleration, the corresponding state transition matrix $\boldsymbol{\Phi}$ and noise driven matrix $\boldsymbol{\Gamma}$ are

$$\boldsymbol{\Phi} = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \boldsymbol{\Gamma} = \begin{bmatrix} \frac{T^2}{2} & 0 \\ 0 & \frac{T^2}{2} \\ T & 0 \\ 0 & T \end{bmatrix} \tag{31}$$

Algorithm 1 DAPA-EKF, LS-AEKF, and LS-VEKF

State equation: $\boldsymbol{X}(k+1) = \boldsymbol{\Phi}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{W}(k)$, where $\boldsymbol{X}(k)$ and $\boldsymbol{W}(k)$ are the state and white noise at discrete time k, respectively. The state transition matrix $\boldsymbol{\Phi}$ and noise driven matrix $\boldsymbol{\Gamma}$ are given in (15) for DAPA-EKF and LS-AEKF and in (31) for LS-VEKF.

Observation equation: $\boldsymbol{Z}(k) = \boldsymbol{H}(k)\boldsymbol{X}(k) + \boldsymbol{V}(k)$, where $\boldsymbol{V}(k)$ is the observation noise. The observation matrix $\boldsymbol{H}(k)$ is given by (18), (22), and (29) for DAPA-EKF (LOS = 1), DAPA-EKF (LOS = 2), and LS-AEKF/LS-VEKF, respectively.

1: Noise covariance matrix: $\boldsymbol{Q} = \boldsymbol{E}\left(\boldsymbol{W}_j^2\right)$, $\boldsymbol{R} = \boldsymbol{E}\left(\boldsymbol{V}_j^2\right)$.

2: State prediction:

$$\hat{\boldsymbol{X}}\left(k+1\middle|k\right) = \boldsymbol{\Phi}\hat{\boldsymbol{X}}\left(k\middle|k\right). \tag{32}$$

3: Covariance matrix prediction:

$$\boldsymbol{P}\left(k+1\middle|k\right) = \boldsymbol{\Phi}\boldsymbol{P}\left(k\middle|k\right)\boldsymbol{\Phi}^T + \boldsymbol{\Gamma}\boldsymbol{Q}\boldsymbol{\Gamma}^T. \tag{33}$$

4: Kalman gain:

$$\boldsymbol{K}(k+1) = \boldsymbol{P}\left(k+1\middle|k\right)\boldsymbol{H}^T\left[\boldsymbol{H}\boldsymbol{P}\left(k+1\middle|k\right)\boldsymbol{H}^T + \boldsymbol{R}\right]^{-1}. \tag{34}$$

5: Status update:

$$\hat{\boldsymbol{X}}\left(k+1\middle|k+1\right) = \hat{\boldsymbol{X}}\left(k+1\middle|k\right) + \boldsymbol{K}(k+1)\boldsymbol{\varepsilon}(k+1), \tag{35}$$

$$\boldsymbol{\varepsilon}(k+1) = \boldsymbol{Z}(k+1) - \boldsymbol{H}\hat{\boldsymbol{X}}\left(k+1\middle|k\right). \tag{36}$$

6: Covariance update:

$$\boldsymbol{P}\left(k+1\middle|k+1\right) = [\boldsymbol{I}_n - \boldsymbol{K}(k+1)\boldsymbol{H}]\boldsymbol{P}\left(k+1\middle|k\right),$$

$$\hat{\boldsymbol{X}}\left(0\middle|0\right) = \mu_0, \boldsymbol{P}\left(0\middle|0\right) = P_0. \tag{37}$$

respectively. Taking the same observation matrix and vector as LS-AEKF, the LS-VEKF can be obtained by filtering the predicted position using EKF.

D. TDOA-KF Positioning Algorithm for LOS ≥ 3

In addition, to further improve the robustness of positioning scheme, a TDOA-KF is proposed based on both the TDOA and the KF. The positioning accuracy of TDOA algorithm is measured by geometric dilution of precision (GDOP), which is defined as the distance vector amplification factor of the ranging error [46]. The positioning error or time error can be obtained by multiplying the ranging error by an appropriate GDOP, which can usually be obtained by the navigation covariance matrix. The principle of the TDOA algorithm is shown in Fig. 10, where three BSs can be used for positioning, which corresponds to a hyperbolic positioning algorithm. The time and distance from tag (x, y) to BS1(x_1, y_1), BS2(x_2, y_2), and BS3(x_3, y_3) are $t_i, r_i (i = 1, 2, 3)$, and the distance difference is $r_{i,j}$, i.e.,

$$r_{ij} = r_i - r_j = v \times (t_1 - t_2) \tag{38}$$

$$r_i = r_{i,1} + r_1. \tag{39}$$

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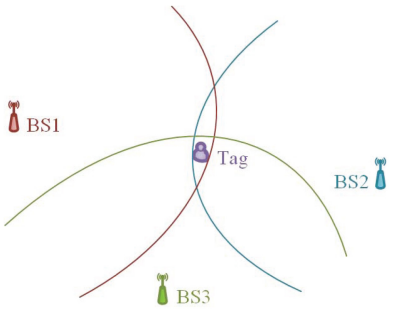


Fig. 10. TDOA positioning diagram.

Therefore, the hyperbola equation is established and the tag coordinates are obtained as

$$\begin{cases} r_{12} = \sqrt{(x_1-x)^2+(y_1-y)^2} - \sqrt{(x_2-x)^2+(y_2-y)^2} \\ r_{13} = \sqrt{(x_1-x)^2+(y_1-y)^2} - \sqrt{(x_3-x)^2+(y_3-y)^2}. \end{cases} \tag{40}$$

Based on 38, we have

$$r_{i,1}^2 + 2r_{i,1}r_1 + r_1^2 = K_i - 2x_i x - 2y_i y + x^2 + y^2 \tag{41}$$

$$K_i = x_i^2 + y_i^2. \tag{42}$$

If $i = 1$, (39) is equivalent to

$$r_1^2 = K_1 - 2x_1 x - 2y_1 y + x^2 + y^2. \tag{43}$$

Equation (41) subtracted from (39) yields

$$x_{i,1}x + y_{i,1}y = -r_{i,1}r_1 + \frac{K_i - K_1 - r_{i,1}^2}{2}. \tag{44}$$

When four BSs are evenly distributed in the indoor environment, the positioning error of TDOA algorithm is defined as

$$r_{i,1} = \sqrt{(x-x_i)^2+(y-y_i)^2} - \sqrt{(x-x_1)^2+(y-y_1)^2}. \tag{45}$$

To derive the distance error and station site error, we first derive

$$\begin{aligned} dr_{i,1} &= \frac{(x-x_i)dx + (x-x_i)dx_i + (y-y_i)dy + (y-y_i)dy_i}{\sqrt{(x-x_i)^2+(y-y_i)^2}} \\ &\quad - \frac{(x-x_1)dx + (x-x_1)dx_1 + (y-y_1)dy + (y-y_1)dy_1}{\sqrt{(x-x_1)^2+(y-y_1)^2}} \\ &= \left(\frac{x-x_i}{r_i} - \frac{x-x_1}{r_1}\right)dx + \left(\frac{y-y_i}{r_i} - \frac{y-y_1}{r_1}\right)dy + (k_i - k_1) \end{aligned} \tag{46}$$

where

$$k_i = \frac{x-x_i}{r_i}dx_i + \frac{y-y_i}{r_i}dy_i. \tag{47}$$

Then, the distance error can be figured out as

$$d\boldsymbol{R} = \boldsymbol{H}d\boldsymbol{X} + d\boldsymbol{S} \tag{48}$$

where

$$d\boldsymbol{R} = \begin{bmatrix} dr_{2,1} & dr_{3,1} & dr_{4,1} \end{bmatrix}^T \tag{49}$$

$$\boldsymbol{H} = \begin{bmatrix} \frac{x-x_2}{r_i} - \frac{x-x_1}{r_1} & \frac{y-y_2}{r_i} - \frac{y-y_1}{r_1} \\ \frac{x-x_3}{r_i} - \frac{x-x_1}{r_1} & \frac{y-y_3}{r_i} - \frac{y-y_1}{r_1} \\ \frac{x-x_4}{r_i} - \frac{x-x_1}{r_1} & \frac{y-y_4}{r_i} - \frac{y-y_1}{r_1} \end{bmatrix}, d\boldsymbol{X} = \begin{bmatrix} dx \\ dy \end{bmatrix}. \tag{50}$$

$$d\boldsymbol{S} = \begin{bmatrix} k_2 - k_1 & k_3 - k_1 & k_4 - k_1 \end{bmatrix}^T. \tag{51}$$

Eventually, the positioning error $d\boldsymbol{X}$ can be determined by using the distance error, $d\boldsymbol{R}$, and station site error, $d\boldsymbol{S}$, i.e.,

$$d\boldsymbol{X} = \boldsymbol{H}^{-1}(d\boldsymbol{R} - d\boldsymbol{S}) = \left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T(d\boldsymbol{R} - d\boldsymbol{S}). \tag{52}$$

The covariance matrix of positioning error is

$$\begin{aligned} &\operatorname{cov}(d\boldsymbol{X}) = E\left[d\boldsymbol{X}d\boldsymbol{X}^T\right] \\ &= E\left[\left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T(d\boldsymbol{R} - d\boldsymbol{S})\left[\left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T(d\boldsymbol{R} - d\boldsymbol{S})\right]^T\right] \\ &= E\left[\left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T(d\boldsymbol{R} - d\boldsymbol{S})\left[\left(d\boldsymbol{R}^T - d\boldsymbol{S}^T\right)\left(\boldsymbol{H}\left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\right)\right]^{-1}\right] \\ &= \left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T E\left[\left(d\boldsymbol{R}d\boldsymbol{R}^T\right) + \left(d\boldsymbol{S}d\boldsymbol{S}^T\right)\right]\boldsymbol{H}\left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1} \\ &= \left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T[\operatorname{cov}(d\boldsymbol{R}) + \operatorname{cov}(d\boldsymbol{S})]\boldsymbol{H}\left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}. \end{aligned} \tag{53}$$

It should be noticed that the distance error and the station site error are independent of each other, leading to

$$E\left[d\boldsymbol{R}d\boldsymbol{S}^T\right] = E\left[d\boldsymbol{S}d\boldsymbol{R}^T\right] = 0. \tag{54}$$

If $d\boldsymbol{R}$ follows a normal distribution as $N(0, \sigma_{r_{i,1}})$, the correlation coefficient between $d\Delta r_{i,1}$ and $d\Delta r_{j,1}$ can be given by

$$\eta_{i,j} = \frac{\operatorname{cov}\left(dr_{i,1}, dr_{j,1}\right)}{\sigma_{r_{i,1}}\sigma_{r_{j,1}}} \tag{55}$$

where the covariance matrix, $\operatorname{cov}(dr_{i,1}, dr_{j,1})$, is

$$\begin{bmatrix} \sigma_{r_{2,1}}^2 & \eta_{23}\sigma r_{2,1}\sigma r_{3,1} & \eta_{24}\sigma r_{2,1}\sigma r_{4,1} \\ \eta_{32}\sigma r_{3,1}\sigma r_{2,1} & \sigma_{r_{3,1}}^2 & \eta_{34}\sigma r_{3,1}\sigma r_{4,1} \\ \eta_{42}\sigma r_{4,1}\sigma r_{2,1} & \eta_{43}\sigma r_{4,1}\sigma r_{3,1} & \sigma_{r_{4,1}}^2 \end{bmatrix}. \tag{56}$$

The station site error is distributed independently in horizontal and vertical directions following $N(0, \sigma_s)$. Thus, we have

$$\begin{aligned} &\operatorname{cov}\left(d\boldsymbol{S}_i, d\boldsymbol{S}_j\right) = E\left[(k_i - k_1)\left(k_j - k_1\right)\right] \\ &= E\left[\left[\left(\frac{x-x_i}{r_i}dx_i + \frac{y-y_i}{r_i}dy_i\right) - \left(\frac{x-x_1}{r_1}dx_1 + \frac{y-y_1}{r_1}dy_1\right)\right]\right. \\ &\qquad \left.\times\left[\left(\frac{x-x_i}{r_i}dx_i + \frac{y-y_i}{r_i}dy_i\right) - \left(\frac{x-x_1}{r_1}dx_1 + \frac{y-y_1}{r_1}dy_1\right)\right]\right] \\ &= E\left[\left[\left(e_{x_i}dx_i + e_{y_i}dy_i\right) - \left(e_{x_1}dx_1 + e_{y_1}dy_1\right)\right]\right. \\ &\qquad \left.\cdot\left[\left(e_{x_j}dx_j + e_{y_j}dy_j\right) - \left(e_{x_1}dx_1 + e_{y_1}dy_1\right)\right]\right] \\ &= \begin{cases} 2\sigma_s^2 & i = j \\ \sigma_s^2 & i \neq j. \end{cases} \end{aligned} \tag{57}$$

Each element f_{ij} in $\operatorname{cov}(d\boldsymbol{R}) + \operatorname{cov}(d\boldsymbol{S}) = [f_{ij}]$ is

$$\begin{cases} \sigma_{ri}^2 + 2\sigma_s^2 & i = j \\ \eta_{i,j}\sigma_{ri}\sigma_{rj} & i \neq j. \end{cases} \tag{58}$$

The covariance is

$$\boldsymbol{P} = \operatorname{cov}(d\boldsymbol{X}) = \left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T[f_{ij}]\left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}. \tag{59}$$

Based on the TDOA algorithm, there are differences in positioning accuracy in the positioning area, which can be expressed by the geometric accuracy of positioning

$$\text{GDOP} = \sqrt{\boldsymbol{P}(1,1) + \boldsymbol{P}(2,2)}. \tag{60}$$

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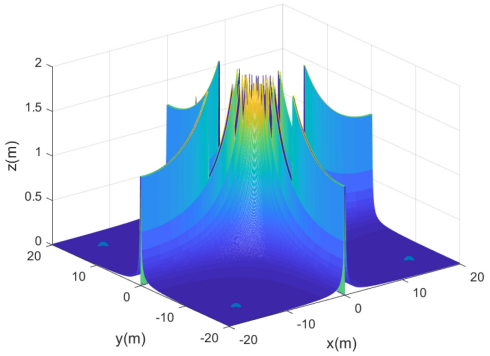


Fig. 11. 3-D illustration of GDOP positioning error.

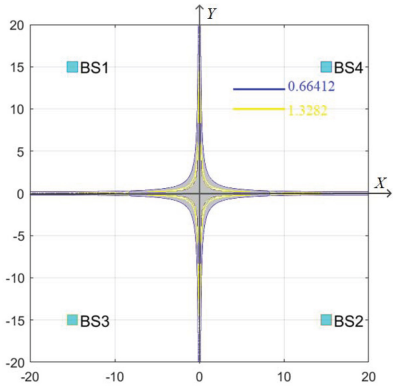


Fig. 12. Error contour map of the TDOA model.

To illustrate the error distribution of the TDOA model, we simulate the model and its positioning error when the BSs are distributed close to the four corners of a square space. The side length of this square space is 40 m centering at (0, 0), and the four BSs are placed at BS1$(-15, 15)$, BS2$(15, -15)$, BS3$(-15, -15)$, and BS4$(15, 15)$. By setting time difference error as 10 ns, the station address error as 0.001 m, the correlation coefficient equals 0, and the positioning error of in this square space is presented in Fig. 11.

The error distribution contour line is shown in Fig. 12, where the corresponding error coefficient is relatively high in the center cross region (marked with gray). In this gray region, the positioning of the tag suffers from relatively larger error. Therefore, LS-AEKF is more appropriate to perform the positioning in this region. In the nongray region, to further improve the positioning accuracy, the positioning results calculated by the TDOA algorithm is subtracted from that calculated by the IMU algorithm as an observation vector. Set state vector $\boldsymbol{X}(k) = [\Delta\hat{x}(k) \quad \Delta\hat{y}(k) \quad \Delta\hat{v_x}(k) \quad \Delta\hat{v_y}(k)]^T$, which represents the difference between UWB and IMU predictions. Therefore, the observation equation can be expressed as

$$\boldsymbol{Z}(k) = \boldsymbol{H}\boldsymbol{X}(k) + \boldsymbol{V}(k). \tag{61}$$

$\boldsymbol{V}(k) = \begin{bmatrix} v_x(k) & v_y(k) & v_{v_x}(k) & v_{v_y}(k) \end{bmatrix}^T$ with a mean value of zero covariance matrix $\boldsymbol{R} = \text{diag}(\sigma_x, \sigma_y, \sigma_{v_x}, \sigma_{v_y})$. The observation vector and the observation matrix are shown in (60) and (61), respectively, that are

$$\boldsymbol{Z}(k) = \begin{bmatrix} x_k^{\text{UWB}} - x_k^{\text{IMU}} \\ y_k^{\text{UWB}} - y_k^{\text{IMU}} \\ v_{x,k}^{\text{UWB}} - v_{x,k}^{\text{IMU}} \\ v_{y,k}^{\text{UWB}} - v_{y,k}^{\text{IMU}} \end{bmatrix} \tag{62}$$

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{63}$$

The state matrix and covariance matrix are updated based on the observation vector and observation matrix by KF algorithm flow.

IV. Simulation and Experiment

In this part, we evaluate the performance of the proposed adaptive fusion positioning algorithm through both simulations and lab experiments. In the simulation, with the assumption that the motion is linear and uniformly accelerated, the tag is initialized at the reference coordinate (0, 0). The initial velocities in the horizontal and vertical directions are set to 0.15 m/s and the acceleration is set to 0.002 m/s^2. During the motion, a process noise with a small variance of $Q = 10^{-8}$ is added to the trajectory. Then, based on this motion model, observations are generated under a certain observation noise variance R. The signal sampling period is $t = 1$ s, and the total simulation time is $N = 50$ s.

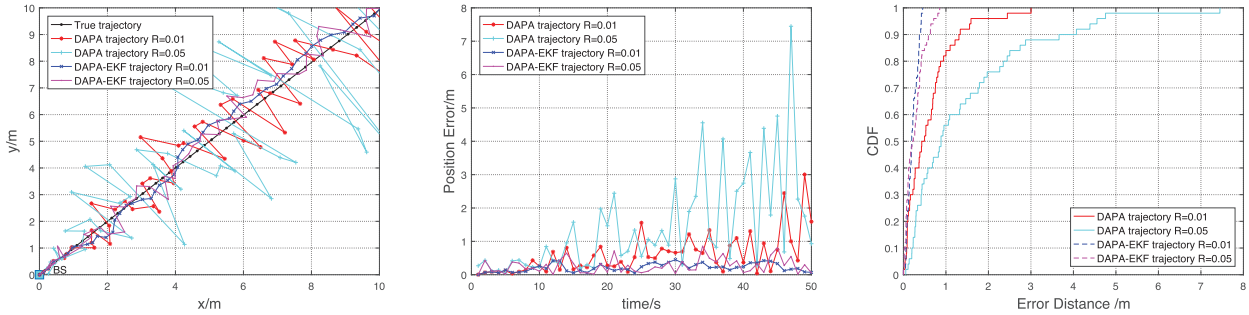
A. LOS = 1

When LOS = 1 and the BS is located at the (0, 0) position, we consider two scenarios where the observed noise variances are $R = 0.01$ and $R = 0.05$. The proposed adaptive fusion method selects DAPA-EFK in the scenario. The simulation results of both the DAPA-EKF method and a reference DAPA method under the two scenarios are plotted in Fig. 13(a). As shown in the figure, the reference DAPA method suffers from large fluctuation for both $R = 0.01$ and $R = 0.05$ scenarios. In comparison, the DAPA-EKF effectively reduces the positioning fluctuation achieving highly accurate performance, but the jittering of positioning increases with time. Fig. 13(b) shows that the observation noise adversely affects the positioning accuracy of DAPA, particularly for the $R = 0.05$ scenario. With the EKF filtering, the proposed DAPA-EKF method is relatively robust against the observation noise. Therefore, as shown in Fig. 13(c), the CDFs of the DAPA-EKF have maximum positioning errors less than 1 m for both $R = 0.01$ and $R = 0.05$ scenarios. In comparison, the CDFs of the reference DAPA have maximum errors of 3 and 7.5 m for $R = 0.01$ and $R = 0.05$, respectively. The results show that the positioning jitter of a single BS will increases with time.

B. LOS = 2

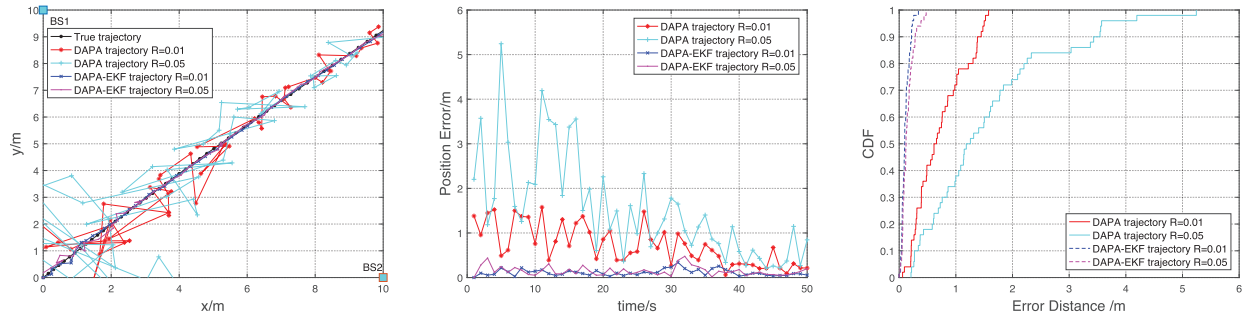
When LOS = 2 and the BSs are located at the (0, 10) and (10, 0) positions, we consider two scenarios where the observed noise variances are $R = 0.01$ and $R = 0.05$. The

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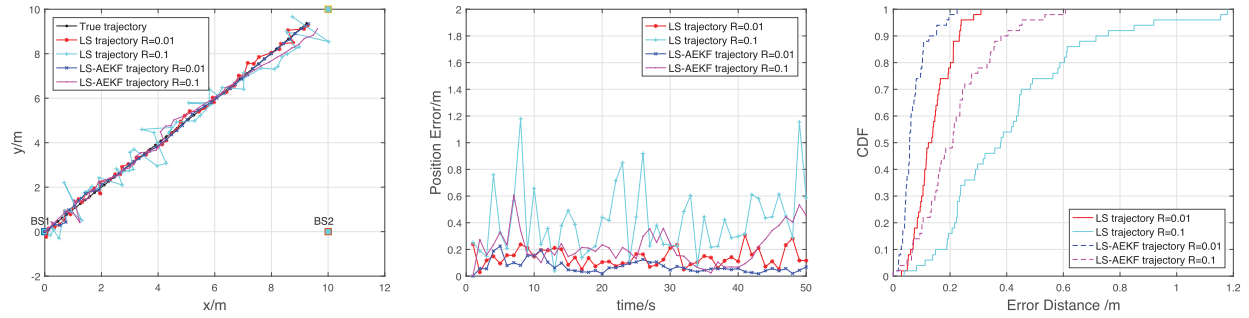
(a) (b) (c)

Fig. 13. Performance of the proposed DAPA-EKF algorithm for LOS = 1. (a) One BS DAPA-EKF trajectory. (b) Ranging error. (c) CDF of error.



(a) (b) (c)

Fig. 14. Performance of the proposed DAPA-EKF algorithm for LOS = 2. (a) Two BSs DAPA-EKF trajectory. (b) Ranging error. (c) CDF of error.



(a) (b) (c)

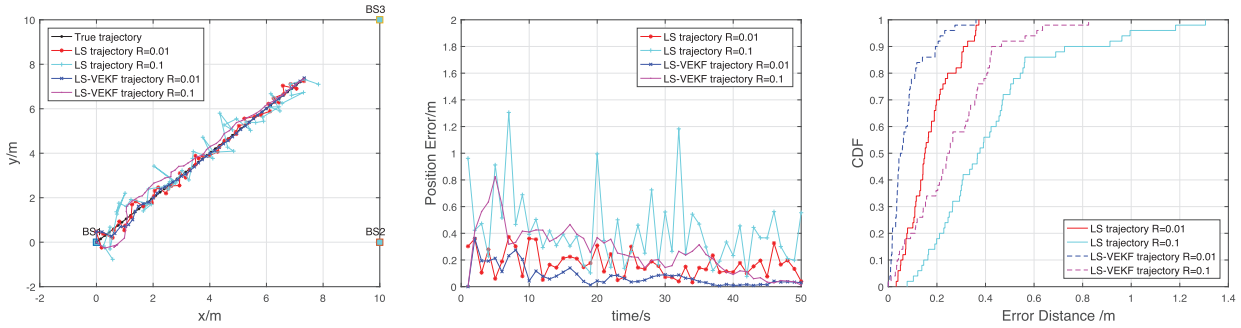
Fig. 15. Performance of the proposed LS-AEKF algorithm for LOS = 3. (a) LS-AEKF trajectory. (b) Ranging error. (c) CDF of error.

proposed adaptive fusion method selects DAPA-EFK in the scenario. The simulation results of both the DAPA-EKF method and a reference DAPA method under the two scenarios are plotted in Fig. 14(a). Compared with the scenarios when only one BS is available, the reference DAPA method based on two BSs is less sensitive to the observation noise. In contrast to the single BS scenario, the DAPA error in the simulation of two BSs is large at the beginning and decreases with time. Fig. 14(b) shows that the error of both the proposed and the reference methods increase with the observation noise. The CDFs of both methods with different observation noises are shown in Fig. 14(c). The maximum positioning error of DAPA-EKF is within 0.5 m for both $R = 0.01$ and $R = 0.05$ scenarios. When the observation noise variance is $R = 0.01$, the maximum error of the reference DAPA positioning is about 1.5 m. When the observation noise variance is $R = 0.05$, the maximum error of the reference DAPA positioning is close to 5.2 m.

C. LOS ≥ 3

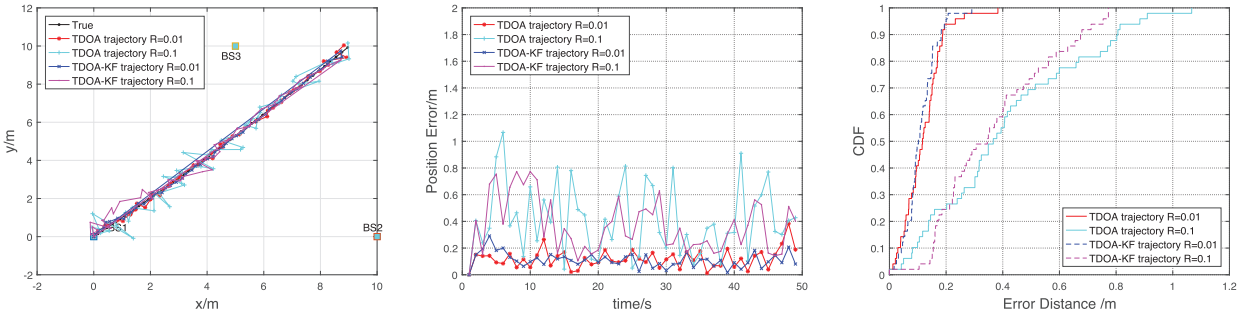
When LOS = 3 and the BSs are located at the (0, 0), (0, 10), and (10, 10) positions, we consider two scenarios where the observed noise variances are $R = 0.01$ and $R = 0.1$. The proposed adaptive fusion method selects LS-AEKF in this scenario. The simulation results of both the LS-AEKF method and a reference LS method under the two scenarios are plotted in Fig. 15(a). The positioning errors of the two positioning algorithms under different observation noises are shown in Fig. 15(b). The observation noise has a great impact on the positioning results of the reference LS method, while the proposed LS-AEKF effectively reduces the jitter of

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(a) (b) (c)

Fig. 16. Performance of the proposed LS-VEKF algorithm for LOS = 3. (a) LS-VEKF trajectory. (b) Ranging error. (c) CDF of error.



(a) (b) (c)

Fig. 17. Performance of the proposed TDOA-KF algorithm for LOS = 3. (a) TDOA-KF trajectory. (b) Ranging error. (c) CDF of error.

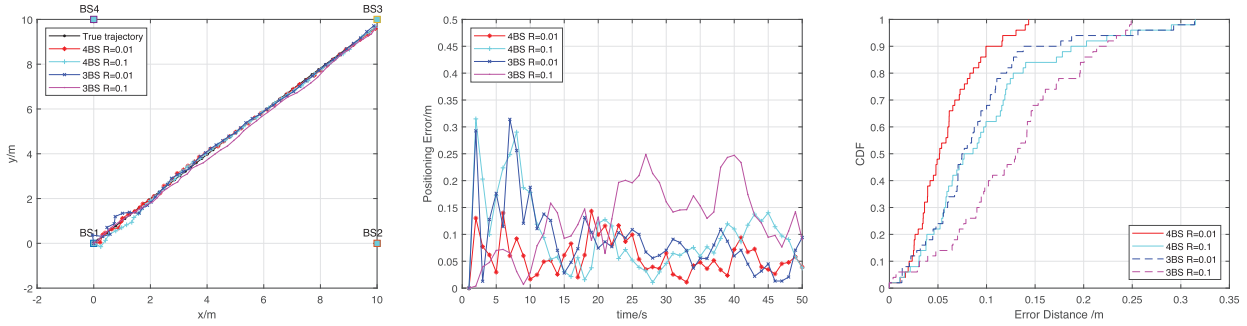
positioning data. The CDFs of both methods are shown in Fig. 15(c). When the observation noise variance is $R = 0.1$, the maximum positioning error of the reference LS is close to 1.2 m while that of LS-AEKF is about 0.6 m. When the observation noise variance is $R = 0.01$, the maximum positioning error of LS is about 0.3 m while that of LS-AEKF is about 0.2 m. In addition to LS-AEKF, we also evaluate the performance of LS-VEKF against the reference LS method. Similarly, the trajectories of the reference LS and the LS-VEKF under the observation noise variances of $R = 0.1$ and $R = 0.01$ are plotted in Fig. 16(a). The positioning errors and the corresponding CDFs are shown in Fig. 16(b) and (c), respectively. When the observation noise variance are $R = 0.1$, the maximum positioning error of LS is about 1.3 m while that of LS-AEKF is about 0.8 m. When the observation noise variance is $R = 0.01$, the maximum positioning error of LS is about 0.4 m while that of LS-VEKF is about 0.4 m. However, as can be seen from the CDF figure, the positioning error of LS-VEKF at this time is 90% maintained at 0.2 m, while that of LS is only 70% maintained at 0.2 m. Fig. 17(a) shows the positioning trajectories of TDOA and TDOA-KF under the observation noise variances are $R = 0.1$ and $R = 0.01$. As shown in Fig. 17(b), TDOA-KF has an overall smaller error than TDOA. The CDF is shown in Fig. 17(c). When the observation noise variance is $R = 0.1$, the maximum positioning errors of TDOA-KF and TODA are approximately 0.8 and 1.07 m, respectively. When the observation noise variance is $R = 0.01$, the maximum positioning errors of TDOA-KF and TODA are approximately 0.2 and 0.4 m, respectively. Therefore, the KF can significantly alleviate the measurement error of time difference and improve the positioning accuracy.

TABLE I
COMPARISON OF MEAN ERROR OR STANDARD DEVIATION OF DIFFERENT ALGORITHMS WITH THREE BSS

Observation noise	Algorithm	Mean error	Standard deviation
R=0.1	LS	0.7364 m	0.7692
R=0.01	LS	0.1693 m	0.0864
R=0.1	LS-VEKF	0.1414 m	0.4494
R=0.01	LS-VEKF	0.0373 m	0.0648
R=0.1	LS-AEKF	0.1206 m	0.3637
R=0.01	LS-AEKF	0.0124 m	0.0233
R=0.1	TDOA	0.7294 m	0.5050
R=0.01	TDOA	0.4076 m	0.0791
R=0.1	TDOA-KF	0.7217 m	0.5932
R=0.01	TDOA-KF	0.0576 m	0.0340

When LOS = 3, the comparison of mean error and standard deviation of each algorithm is shown in Table I. It can be seen that observation noise is an important factor that affects positioning accuracy. Furthermore, we conclude that the improved LS and TDOA algorithms reduce the positioning error, and LS-AEKF has the smallest positioning mean error and standard deviation compared with that of LS-VEKF and TDOA-KF. The proposed algorithms, i.e., LS-AEKF, LS-VEKF, and TDOA-KF, are valid for both cases of LOS = 3 and LOS = 4. Taking an example of the case LOS = 3, Figs. 15–17 show that the proposed algorithms outperform the

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(a) (b) (c)

Fig. 18. Performance of the proposed LS-AEKF algorithm for LOS = 3 or 4. (a) LS-AEKF trajectory. (b) Ranging error. (c) CDF of error.



Fig. 19. Experimental environment.

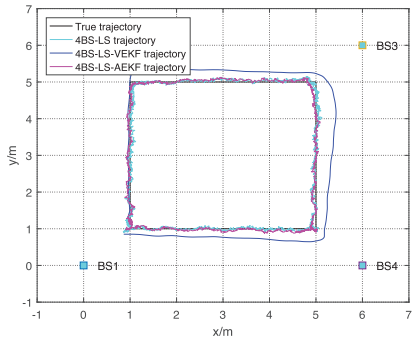


Fig. 20. Actual and predicted trajectories of tag during the positioning experiment for LOS = 4.

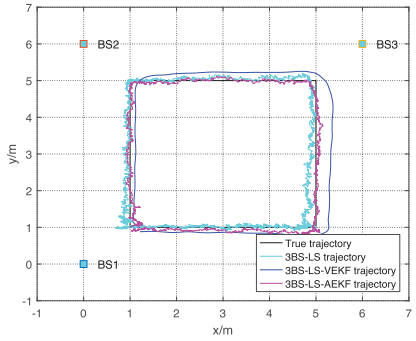


Fig. 21. Actual and predicted trajectories of tag during the positioning experiment for LOS = 3.

corresponding baseline methods. Furthermore, Fig. 18 shows that the positioning performance of four BSs is better than that of three BSs when the noise level is the same. This is because more BSs bring about more distance data in space, from which more accurate positioning information can be extracted. The achieved fairly good positioning accuracy demonstrates the effectiveness of the proposed positioning algorithm as well as the NLOS identification approach.

D. Experiments and Results

In this work, we employ the DW1000 UWB system, which conforms to the IEEE 802.15.4-2011 UWB standard. It supports six radio frequency bands from 3.5 to 6.5 GHz, and can be used in either TWR or TDOA positioning system. DW1000 consists of an analog front end, a receiving end or Tx/Rx switch transmitter, and temperature and voltage monitors. The analog frond end, which is connected to the main processor outside the chip, includes a receiver, a transmitter, and a digital back end. The receiving end connects to the antenna port. The temperature and voltage monitors are provided on the chip. The maximum packet length for high throughput applications is 1023 bytes.

The experiment is performed in a square lab with 6 m $\times$ 6 m size as shown in Fig. 19. There are four BSs placed in the four corners of the space. The tag moves along the evenly divided grid line, and each grid is 1 m $\times$ 1 m square.

In the experiment with four BSs available, we select and evaluate three methods: 1) the LS; 2) LS-VEKF; and 3) LS-AEKF algorithms. The actual trajectory and the predicted trajectories of the three methods are plotted in Fig. 20. It can be seen from the figure that it is necessary to consider the effect of acceleration in an indoor environment with low-speed movement. The LS-AEKF method fits the actual trajectory well with small standard deviation of the positioning error, and the jittering of positioning is smaller than that of LS. Since the LS-VEKF algorithm does not consider the influence of acceleration, the movement trajectory will be deviated due to the deviation of the direction angle in the practical application scenario. Similarly, when only three BSs are available, the positioning trajectories of the three methods are shown in Fig. 21. For both four and three BSs scenarios, the result of positioning error is shown in Fig. 22. The proposed LS-AEKF has higher positioning accuracy than LS-VEKF. Therefore, LS-AEKF is selected

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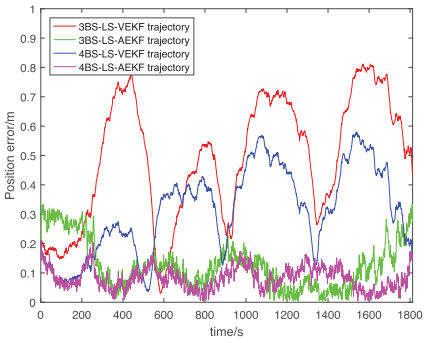


Fig. 22. Positioning error for LOS = 3 and 4.

TABLE II
COMPARISON OF MEAN ERROR OR STANDARD DEVIATION BETWEEN LS-VEKF AND LS-AEKF

Algorithm	Mean error	Standard deviation
3 BSs LS-VEKF	0.4902 m	0.2105
3 BSs LS-AEKF	0.1269 m	0.0819
4 BSs LS-VEKF	0.3129 m	0.1470
4 BSs LS-AEKF	0.0821 m	0.0374

by our adaptive fusion method for LOS $\geq$ 3 situations. The positioning performance in terms of mean error and standard deviation is presented in Table II for LOS $\geq$ 3 scenarios. It can be observed that the positioning accuracy of the proposed LS-AEKF method is 73.8%–74.1% higher than a reference LS-VEKF method in terms of three or four BSs metrics. From the positioning error distribution, it can be seen that the change of the movement direction will cause the IMU positioning distortion, and EKF can effectively alleviate the error caused by the deviation of the acceleration direction.

V. CONCLUSION

In this article, a high-precision adaptive fusion method is proposed for UWB/IMU-based systems to perform indoor positioning under NLOS situation. This fusion method employs an SVM model to detect NLOS situation and the number of LOS BS. Then, based on the number of LOS BSs, the proposed fusion method adaptively applies the most appropriate strategy to maximize the positioning accuracy. In particular, when there are one or two LOS BSs, DAPA-EKF fusion algorithm is adopted; when there are three or more LOS BSs are available, the LS-AEKF algorithm is adopted. The experiment is performed using four BSs in a square indoor testing scenario. Validated using more than 1000 points, the results show that the proposed method significantly outperforms the reference methods under various NLOS situations. DAPA-EKF fusion algorithm is ulilized if the number of available LOS BS is less than three and LS-AEKF algorithm is leveraged otherwise.

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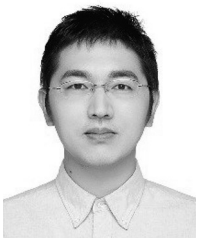
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Daquan Feng (Member, IEEE) received the Ph.D. degree in information engineering from the National Key Laboratory of Science and Technology on Communications, University of Electronic Science and Technology of China, Chengdu, China, in 2015.

He was a visiting student with the School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, USA, from 2011 to 2014. After graduation, he was a Research Staff with the State Radio Monitoring Center, Beijing, China, and then a Postdoctoral Research Fellow with Singapore University of Technology and Design, Singapore. He is currently an Associate Professor with the Shenzhen Key Laboratory of Digital Creative Technology, the Guangdong Province Engineering Laboratory for Digital Creative Technology, the Guangdong–Hong Kong Joint Laboratory for Big Data Imaging and Communication, College of Electronics and Information Engineering, Shenzhen University, Shenzhen, China. His research interests include URLLC communications, MEC, and massive IoT networks.

Dr. Feng is an Associate Editor of IEEE COMMUNICATIONS LETTERS, *ICT Express*, and *Digital Communications and Networks*.



Junjie Peng received the master's degree in electronics and communication engineering from the School of Electronics and Information Engineering, Shenzhen University, Shenzhen, China, in 2022.

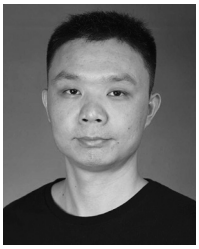
After graduation, he worked with West Kowloon Station Police Station, Shenzhen Department, Guangzhou Railway Public Security Bureau, Shenzhen.



Yuan Zhuang (Senior Member, IEEE) received the bachelor's degree in information engineering and the master's degree in microelectronics and solid-state electronics from Southeast University, Nanjing, China, in 2008 and 2011, respectively, and the Ph.D. degree in geomatics engineering from the University of Calgary, Calgary, AB, Canada, in 2015.

He is a Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China. To date, he has coauthored over 100 academic papers and 20 patents and has received over ten academic awards. His current research interests include multisensors integration, real-time location system, wireless positioning, Internet of Things, and machine learning for navigation applications.

Dr. Zhuang is an Associate Editor of IEEE ACCESS, the Guest Editor of the IEEE INTERNET OF THINGS JOURNAL and *Satellite Navigaiton*, and a reviewer of over 20 IEEE journals.



Chongtao Guo (Member, IEEE) received the B.Eng. and Ph.D. degrees from Xidian University, Xi'an, China, in 2009 and 2014, respectively.

From December 2017 to 2018, he was a Postdoctoral Research Fellow with Georgia Institute of Technology, Atlanta, GA, USA. Since 2014, he has been with Shenzhen University, Shenzhen, China, where he is currently an Associate Professor. His current research interests are in optimization and learning for wireless communications, Internet of Things, and vehicular networks.

Dr. Guo is a co-recipient of the Best Paper Award at the 21st International Conference on Digital Signal Processing (2016 DSP) and the 22nd International Conference on Digital Signal Processing (2017 DSP).

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Tingting Zhang (Member, IEEE) received the B.S. (Hons.), M.S., and Ph.D. degrees in electronic engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2003, 2005, and 2009, respectively.

He is currently a Professor with HIT, Shenzhen, China. His main research interests include wireless localization, integration of sensing and communication, and autonomous vehicle navigation and control.

Prof. Zhang received the Outstanding Postdoctoral Award of HIT, Shenzhen Graduate School, Shenzhen, in 2011. He also received the Shenzhen High Level Talent Program Award in 2012. He serves as an Associate Editor for *Frontiers in Communications and Networks*, a Guest Editor for many international journals, such as *IEEE Communications Magazine*, and also an Organizer and a TPC Member for several international conferences, such as IEEE ICC, GLOBECOM, ICCC, and VTC.



Yinghao Chu received the Ph.D. degree from the University of Califnornia at San Diego, CA, USA, in 2015.

He is currently an Associate Research Professor working with Shenzhen University, Shenzhen, China. He has been working in the domain of artificial intelligence (AI) for real-world application since 2011. Since February 2023, has been with the Department of Advanced Design and Systems Engineering, City University of Hong Kong, Hong Kong. His research focuses on hybrid AI, which simulates the attention and coordination mechanism of human intelligence to solve application-orientated problems in real world, particularly in the areas of 1) renewable forecast and application, such as probabilistic forecasts of solar/load time series and 2) smart manufacturing, such as operational computer vision system for robot guidance and surface quality inspection.



Xiaoan Zhou received the Ph.D. degree in circuits and systems from South China University of Technology, Guangzhou, China, in 2000.

Since July 2000, he has been with the College of Electronics and Information Engineering, Shenzhen University, Shenzhen, China as a Lecturer and then an Associate Professor. His research interests include mobile and wireless communications, cognitive radio, and spectrum sharing technologies. From August 2019 to August 2020, he was a Visiting Scholar with the Department of ECE, University of Waterloo, Waterloo, ON, Canada. His current research interests include 5G, mobile-edge computing, and Internet of Things.

Dr. Zhou won the Second Prize of Guangdong Provincial Science and Technology Award in 2006 and the Shenzhen Innovation Award in 2007, as a key participant.



Xiang-Gen Xia (Fellow, IEEE) received the B.S. degree in mathematics from Nanjing Normal University, Nanjing, China, in 1983, and the M.S. degree in mathematics from Nankai University, Tianjin, China, in 1986, and the Ph.D. degree in electrical engineering from The University of Southern California, Los Angeles, CA, USA, in 1992.

He was a Senior/Research Staff Member of Hughes Research Laboratories, Malibu, CA, USA, from 1995 to 1996. In September 1996, he joined the Department of Electrical and Computer Engineering, University of Delaware, Newark, DE, USA, where he is the Charles Black Evans Professor. He is the author of the book *Modulated Coding for Intersymbol Interference Channels* (New York: Marcel Dekker, 2000). His current research interests include space time coding, MIMO and OFDM systems, digital signal processing, and SAR and ISAR imaging.

Dr. Xia received the National Science Foundation Faculty Early Career Development (CAREER) Program Award in 1997, the Office of Naval Research Young Investigator Award in 1998, and the Outstanding Overseas Young Investigator Award from the National Natural Science Foundation of China in 2001. He received the 2019 Information Theory Outstanding Overseas Chinese Scientist Award of the Information Theory Society of Chinese Institute of Electronics. He has served as an Associate Editor for numerous international journals, including IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON MOBILE COMPUTING, and IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY. He is a Technical Program Chair of the Signal Processing Symposium, Globecom 2007 in Washington, DC., USA, and the General Co-Chair of ICASSP 2005 in Philadelphia.

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